

Programme	B. Sc. Computer Science					
Course Title	System Security					
Type of Course	Major					
Semester	VII					
Academic Level	400 - 499					
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours	
	4	4	-	-	60	
Pre-requisites	Knowledge in Fundamentals of Network and Operating System and COA					
Course Summary	The syllabus is prepared with the view of preparing the BSc Computer Science Graduates to build effective an understanding of the differences between various forms of computer system security, where they arise, and appropriate tools to achieve them					

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the different types of securities in information and computer systems, security goals and confidentiality, integrity, availability	U	C	Instructor-created exams / Quiz
CO2	Outline computer system threats and various types of system attacks	U	C	Instructor-created exams / Quiz
CO3	Identify different issues associated with system attacks and how attacking occurs and various types of attackers	U	P	Instructor-created exams / Quiz
CO4	Provide knowledge in operating system security, file protections, security assurance	U	C	Instructor-created exams / Case studies
CO5	Understand important elements of Database security	U	P	Instructor-created exams / Quiz Case studies
CO6	Define security planning, various types of security policies and risk analysis	U	P	Instructor-created exams / Quiz / Case studies
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Module	Un it	Content	Hrs	Marks
I	Notion of different types of securities		12	15
	1	Information security - computer security - security goals, relation between security, confidentiality, integrity, availability and authorization, vulnerabilities - principles of adequate protection.	3	
	2	Notions of operating security, database security, program security, network security attacks - threats, vulnerabilities and controls.	3	
	3	The kind of problems - interception, interruption, modification, fabrication.	2	
	4	Computer criminals - amateurs, crackers, career criminals.	2	
	5	Methods of defense control, hardware controls, software controls, effectiveness of controls.	2	
II	Program security		12	15
	6	Secure programs - fixing faults, unexpected behaviour, types of flaws.	2	
	7	Non-malicious program errors - buffer overflows, incomplete mediation.	1	
	8	Viruses and other malicious code - kinds of malicious code, how viruses attach, how viruses gain control, prevention,	3	
	9	Control example - the brain virus, the internet worm, web bugs..	3	
	10	Targeted malicious code - trapdoors, Salami attack	1	
III	11	Controls against program threats - development controls, peer reviews, hazard analysis	2	
	Operating system security		12	20
	12	Protected objects and methods of protection - memory address protection - fence, relocation, base/bounds registers, tagged architecture, segmentation, paging.	2	
	13	Control of access to general objects - directory, access control list	2	
	14	File protection mechanism - basics forms of protection, single permissions.	2	

	15	Authentication - authentication basics, password, authentication process challenge - response, biometrics	2	
	16	Trusted operating systems - security policies for operating systems	2	
	17	Models of security - requirement of security systems, multilevel security, access security, limitations of security systems	2	
IV	Database Security		12	20
	18	Security requirements - integrity, confidentiality and availability of database	2	
	19	Reliability and integrity of database	2	
	20	Sensitive data, interface	3	
	21	Multilevel database	2	
	22	Proposals for multilevel database security	3	
V	Open Ended Module		12	
	CASE STUDY: Administrating security			
	Security planning – Contents of a security planning, team members, commitment to a security plan, business continuity plans. Risk analysis – the nature of risk, steps of risk analysis.			

Reference Books:

1. C. P. Pfleeger and S. L. Pfleeger, Security in Computing, 4th Edition, Pearson India, ISBN: 9788131727256.
2. Matt Bishop, Computer Security: Art & Science, 1st Edition, Pearson, ISBN: 0201440997.
3. William Stallings, Cryptography and Network Security: Principles and Practice, 6th Edition, Pearson India, ISBN: 9332518777.
4. Michael E. Whitman and Herbert J. Mattord, Principles of Information Security, 4th Edition, Cengage Learning India Pvt Ltd, ISBN: 8131516458.

Mapping of COs with PSOs and POs :

	PSO1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
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CO 1	-	2	-	-	1	1						
CO 2	-	3	1	-	1	1						
CO 3	-	2	1	-	1	1						
CO 4	-	2	1	-	1	1						
CO 5	-	3	1	-	1	2						
CO 6	-	2	1	1	1	2						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓	✓		✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	